

Carson Slater

CONTACT INFORMATION	Department of Statistical Science Baylor University One Bear Place 97140 Waco, Texas 76798, USA	Phone (480) 320-8898 E-mail carsonslater7@gmail.com WWW https://carsonslater.github.io
RESEARCH INTERESTS	Time series clustering, anomaly detection, and nonhomogenous hidden markov models with applications in urban water distribution and advanced metering infrastructure technologies.	
EDUCATION	Baylor University PhD Statistical Science Advised by Amanda S. Hering. Research involves developing statistical methods for time series clustering with applications in urban water distribution.	2024–2027 [Expected]
	Baylor University M.S. Statistical Science	2023–2024
	Wheaton College (IL) B.S. Applied Mathematics with Statistics B.A. Economics Edman Presidential Scholarship	2019–2023
PROFESSIONAL EXPERIENCE	Baylor University Research Assistant Developing statistical methods for urban water distribution processes, leveraging unsupervised learning on time series for customer-level water utility data. Research is funded by Denver Water.	2024–Present
	Baylor University Graduate Assistant Consulted for ongoing interdisciplinary research in the computer engineering and environmental science departments, and for the American Holistic Nursing Association.	2023–2025
	Wheaton College Mathematics Teaching Assistant Facilitated help sessions for undergraduate students in the topics of Machine Learning and Probability Theory.	2022–2023
ARTICLES IN PREPARATION	Carson Slater , Ryan Shepler, Samantha Bryan, and Amanda S. Hering. 2026. “Identifying Outdoor Watering Events with High-Frequency Metering Data to Assess Residential Water Consumption”. <i>In Preparation</i> .	
PEER-REVIEWED PUBLICATIONS	Brittany J. Carnathan, Dinny Stevens, Swarna Shikha, Carson Slater , Nathen Byford, Rodney X. Sturdivant, Kuzy Zarzosa, W. Evan Braswell, and Christie M. Sayes. 2025. “Assessing the Effects of Surface-Stabilized Zero-Valent Iron Nanoparticles on Diverse Bacteria Species Using Complementary Statistical Models”. <i>Journal of Functional Biomaterials</i> 16 (3). https://doi.org/10.3390/jfb16030113	

CONFERENCE
PROCEEDINGS & OTHER
ARTICLES

Jonathan E. Swindell, **Carson Slater**, Samuel Hussey, Charles Baylis, and Robert J. Marks. 2024. “Assessing Interference with Regression Analysis Techniques”. *2024 IEEE Texas Symposium on Wireless and Microwave Circuits and Systems (WMCS)*. <https://doi.org/10.1109/WMCS62019.2024.10619025>

Amanda S. Hering, Douglas Nychka, Luke Durell, **Carson Slater**, Michael Poor, Greg Hamerly, and Tzahi Y. Cath. 2024. “Mo(Wa)²TER Introduction to Data Science Course Material”. *Harvard Dataverse*. <https://doi.org/10.7910/DVN/BZJW2E>

Carson Slater and Ying Li. 2022. “Forecasting Fast-Moving Consumer Goods with Generalized Additive Models”. *2022 University of Illinois Chicago Undergraduate Mathematics Symposium*. (Poster Presentation)

HONORS & AWARDS

Wheaton College Athletics, NCAA 2019-22

Varsity Baseball Player
Practiced baseball six days a week for multiple hours and played over forty-games during the spring semester.

Duke/North Carolina State University Summer Institute for Biostatistics 2022

Participant
Engaged standard biostatistical methods; Hackathon studying myocardial infarction.

Valley Christian High School, Chandler AZ 2019

Christian Character Award
Given to one graduating senior each year that exemplified honorable character in their tenure at the high school.

SERVICE & VOLUNTEER

Wheaton Academy Center for Lifelong Learning 2023

Academic Coach
Mentored students 1-on-1 to sharpen math skills from Algebra I to AP Statistics.

Křesťanská Akademie Mladých (CZ) 2023

Volunteer Intern
Assisting local CB churches by teaching English and sports in Czech Republic schools and camps.

SKILLS

Technology	Docker, Positron ($\approx$ VS Code), Git/Github, reveal.js, R/RStudio, SQL, duckDB, L ^A T _E X.
Python	pandas, numpy, PyTorch (Lightning), plotnine, seaborn, plotly, scikit-learn, Jupyter
R	tidyverse (dplyr, ggplot2, etc.), TensorFlow (Keras), torch, shiny, RMarkdown/Quarto, rstan, rjags, caret, tidymodels, targets (gittargets, proffer, etc.)
Statistics	statistical modeling, hypothesis testing, mixture models, time series, regression, linear mixed models, clustering, classification, deep learning, PCA, random forests, ensemble models (e.g., XGBoost), SVM, Bayesian methods, spatiotemporal modeling